

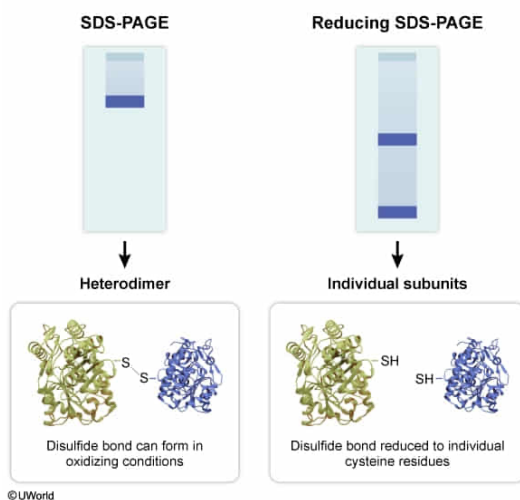
Researchers isolate a protein that runs as a single band on a nonreducing SDS-PAGE gel but as two distinct bands on a reducing gel. The protein is most likely a:

- ✔ ☒ A. heteromultimer that formed in the oxidizing environment of the endoplasmic reticulum. [54%]
- ☐ B. heteromultimer that formed in the reducing environment of the cytosol. [26%]
- ☐ C. homomultimer that formed in the oxidizing environment of the endoplasmic reticulum. [14%]
- ☐ D. homomultimer that formed in the reducing environment of the cytosol. [5%]

Correct

54% answered correctly

Explanation:



Multimeric proteins consist of **two or more polypeptide chains**, called subunits. If all subunits in a multimer are **identical**, the protein is given the prefix **homo-** (homodimer = 2 subunits, homotrimer = 3 subunits, homotetramer = 4 subunits) whereas proteins containing **distinct subunits** are given the prefix **hetero-**.

Multimers are typically bound together by noncovalent interactions but may also have covalent bonds between cysteine residues called **disulfide bonds**. Disulfide bonds form when two cysteine residues are oxidized, and therefore the presence of disulfide bonds requires an oxidizing environment. These bonds can be **broken by reducing agents** such as β -mercaptoethanol or dithiothreitol.

The question states that the protein forms one band in a nonreducing SDS-PAGE gel but in a reducing gel, it forms two bands. The formation of two distinct bands when the multimer is broken suggests that the protein contained nonidentical subunits, making it a **heteromultimer**. The fact that a reducing gel is required to break the multimer indicates that disulfide bonds are present. They most likely formed in the **oxidizing environment** of the **endoplasmic reticulum**.

(Choices B and D) In reducing environments such as the cytosol, proteins are unlikely to form disulfide bonds. Therefore, such proteins are unlikely to be affected by a reducing gel, as they are already reduced.

(Choice C) A homomultimer consists of two or more identical subunits. The individual subunits will run faster on a gel than the intact multimer would, but they will not run differently from each other. Therefore, **only one band** would form on a reducing gel if the protein were a homomultimer.

Educational objective:

Multimeric proteins contain multiple polypeptide subunits. All subunits are identical in homomultimers whereas heteromultimers are composed of different subunits. They may be bound together by disulfide bonds, which must form in oxidizing environments and can only be broken by reducing agents.

Time spent:0 sec

Subject:
Biochemistry

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System:
Amino Acids and Proteins